

BUCURESTI BANEASA, Romania

WMO: 154200

Lat: 44.5106N

Lon: 26.0781E

Elev: 299

StdP: 14.54

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.5	13.8	3.3	6.6	10.5	8.7	8.7	16.2	23.1	30.1	19.5	30.6	4.2	250	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.9	94.0	69.8	91.1	69.2	88.1	68.1	72.9	86.5	71.4	85.0	70.1	83.2	4.9	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.4	109.4	76.5	67.7	102.9	75.0	66.2	97.5	73.9	36.7	86.7	35.4	85.1	34.2	83.1	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.7	14.6	12.2	DB	-1.3	98.2	6.2	3.6	-5.7	100.8	-9.3	102.9	-12.8	104.9	-17.3	107.5	
			WB	-1.7	76.1	6.1	2.5	-6.1	77.8	-9.6	79.3	-13.0	80.7	-17.4	82.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	29.7	34.4	43.0	53.0	62.4	69.7	73.4	73.0	63.8	52.7	43.0	32.8	(6)
	DBStd	16.79	8.46	9.23	8.01	6.47	5.77	5.40	4.64	4.82	6.20	7.13	7.49	8.03	(7)
	HDD50	2175	631	439	240	43	1	0	0	0	54	232	535		(8)
	HDD65	5277	1096	858	682	362	117	19	2	4	94	384	659	1000	(9)
	CDD50	3151	0	2	23	132	387	592	725	713	416	138	23	0	(10)
	CDD65	777	0	0	0	1	38	161	262	252	60	3	0	0	(11)
	CDH74	9455	0	0	2	80	663	1898	3029	2943	765	74	1	0	(12)
	CDH80	3871	0	0	0	13	159	730	1371	1359	229	10	0	0	(13)
(14) Wind	WSAvg	4.8	5.1	5.6	6.0	5.7	4.8	4.3	4.1	3.9	4.3	4.2	4.7	5.2	(14)
(15) Precipitation	PrecAvg	24.8	1.6	1.4	1.6	1.9	2.7	3.0	2.5	2.1	2.1	1.7	1.9	1.9	(15)
	PrecMax	42.4	5.0	3.9	4.4	6.3	9.2	8.1	8.7	5.7	10.6	5.2	8.5	6.7	(16)
	PrecMin	14.7	0.1	0.1	0.1	0.0	0.3	0.4	0.2	0.0	0.2	0.0	0.0	0.0	(17)
	PrecStd	5.6	1.1	0.9	1.1	1.1	1.8	1.8	1.7	1.5	1.9	1.4	1.3	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	64.1	71.7	81.9	88.2	94.3	99.0	98.4	90.9	80.8	69.9	56.4	(19)
		MCWB	45.8	51.8	54.5	61.5	65.7	70.2	70.6	70.3	67.6	63.7	57.4	48.5	(20)
	2%	DB	48.1	57.5	67.1	75.5	84.2	90.8	94.3	94.1	85.9	75.3	64.2	50.5	(21)
		MCWB	42.6	47.6	52.2	58.8	64.0	69.6	70.2	69.6	65.7	60.8	54.3	45.9	(22)
	5%	DB	43.6	52.0	62.7	70.8	80.9	87.5	90.8	91.0	82.2	70.2	58.9	46.4	(23)
		MCWB	41.0	44.8	49.9	56.2	62.9	68.4	69.5	68.8	64.1	58.4	51.9	43.1	(24)
	10%	DB	40.3	46.9	58.3	67.2	77.4	84.2	87.6	87.8	78.5	65.9	54.3	43.0	(25)
		MCWB	38.2	42.5	47.5	54.3	61.4	66.9	68.3	67.8	62.5	56.2	50.2	40.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.8	52.7	55.9	63.0	69.4	74.4	75.1	75.5	70.6	66.1	58.7	50.4	(27)
		MCDB	52.8	61.8	68.3	78.1	83.0	87.8	88.1	88.9	84.3	77.9	67.0	54.1	(28)
	2%	WB	43.6	48.5	53.1	60.2	66.6	71.9	72.9	72.7	67.5	62.5	55.6	46.6	(29)
		MCDB	46.6	56.2	65.1	72.5	79.1	85.5	86.7	86.1	80.4	71.2	61.7	50.2	(30)
	5%	WB	41.1	45.7	50.8	58.0	64.6	70.1	71.5	70.9	65.9	60.0	53.3	43.6	(31)
		MCDB	43.4	50.7	61.2	67.8	76.4	83.1	85.0	84.6	78.0	67.9	58.0	45.8	(32)
	10%	WB	38.1	43.0	48.4	55.7	62.8	68.7	70.1	69.4	64.3	57.6	50.9	40.9	(33)
		MCDB	40.0	46.7	56.2	64.4	74.2	80.9	83.0	82.8	74.8	64.0	53.9	42.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	16.4	20.2	22.6	24.6	24.2	24.9	25.8	23.7	20.8	15.7	12.8	(35)
		MCDBR	18.4	25.9	29.9	30.5	31.3	29.8	30.8	31.7	30.6	28.8	25.0	19.0	(36)
	5% WB	MCWBR	13.1	16.7	16.5	15.1	13.3	11.2	10.7	10.5	12.3	15.1	15.5	13.6	(37)
		MCWBR	17.3	23.5	28.0	26.2	26.7	26.5	26.1	26.7	26.6	25.9	21.4	17.4	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.346	0.378	0.413	0.451	0.427	0.436	0.446	0.461	0.424	0.401	0.374	0.350	(40)
		taud	2.277	2.194	2.155	2.075	2.223	2.231	2.211	2.164	2.233	2.268	2.293	2.291	(41)
	Ebn at Noon	Ebn at Noon	237	250	257	257	268	266	261	252	251	239	224	222	(42)
		Edh at Noon	27	35	42	49	44	43	44	44	38	32	27	24	(43)
(44) All-Sky Solar Radiation	RadAvg	436	701	1089	1492	1862	2011	2037	1817	1317	855	468	360		(44)
	RadStd	67	107	118	153	138	139	133	100	100	104	61	55		(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (62 neighbors)		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	-221	+124	+70	

Nomenclature: See separate page